Phase 2: Data Flow Diagram & User Stories

System Architecture & Agile Specifications

1. Data Flow Architecture (DFD Level 1)

• External Entity (Patient Dataset) -> [P1: Ingestion & Anomaly Audit] -> Data Store (smart_healthcare_dataset.csv)

• [P1] -> [P2: Feature Engineering & Calibration Engine] -> Data Store (smart_healthcare_dataset_cleaned.csv)

• [P2] -> [P3: Tableau Analytics Engine] -> Outputs (smart_healthcare_analysis.twb, project.twbx)

• [P2] -> [P4: Flask REST API Gateway] -> Outputs (Executive Dashboard, Live Patient Explorer, Risk Simulator)

2. User Stories

• US-01: As a clinician, I want to input patient biomarkers and receive an instant calibrated risk score and tier.

• US-02: As an epidemiologist, I want to filter the 5,000-patient cohort by age group and risk tier to study comorbidity.

• US-03: As a college evaluator, I want transparent test verification and a documented explanation of the raw risk score anomaly.